

# TANVIR HOSSAIN

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## RESEARCH INTERESTS

My research develops attacks and defenses for trustworthy microelectronics across the hardware and software stack. I work on power and electromagnetic side-channel analysis, fault injection, hardware Trojan detection and mitigation, and intellectual-property protection, combining digital VLSI and ASIC design flows (RTL design, verification, and synthesis) with machine learning to build golden-chip-free, pre-deployment security evaluation and assurance techniques for integrated circuits.

## EDUCATION

**Doctor of Philosophy (Ph.D.) in Electrical Engineering**, The University of Kansas Expected 2027  
*Advisor: Dr. Tamzidul Hoque, Assistant Professor, EECS, KU.*

**Master of Science (M.S.) in Electrical Engineering**, The University of Kansas May 2025  
*Advisor: Dr. Tamzidul Hoque, Assistant Professor, EECS, KU.*

**Bachelor of Science in Electrical and Electronic Engineering**  
Ahsanullah University of Science and Technology, Bangladesh, CGPA: 3.69 2016 - 2021

## PUBLICATIONS

25 papers total: 18 peer-reviewed publications (8 journal articles, 10 conference papers) and 7 manuscripts under review. Name in **bold**; joint first authorship noted where applicable.

### Journal Articles:

- Tanvir Hossain**, Andrea Ramirez-Salgado, Ertewaa Alsahayan, Swarup Bhunia, Pavlo Antonenko, and Tamzidul Hoque, “**Game On: Gamifying Computer Hardware Fundamentals Learning Using a Configurable Platform**,” *IEEE Transactions on Education*, 2026. DOI: [10.1109/TE.2026.3728763](https://doi.org/10.1109/TE.2026.3728763). (Accepted for publication, forthcoming online)
- Tanvir Hossain**, S. M. Mojahidul Ahsan, Prabuddha Chakraborty, Swarup Bhunia, and Tamzidul Hoque, “**Attacking the Protections of Hardware using Trojan-Assisted Meta Attacks**,” *ACM Transactions on Design Automation of Electronic Systems*, 2026. DOI: [10.1145/3804452](https://doi.org/10.1145/3804452).
- Tanvir Hossain**, Matthew Showers, Mahmudul Hasan, and Tamzidul Hoque, “**On the Security of Lightweight Homomorphic Obfuscation for Protecting Against Hardware Trojans**,” *IACR Transactions on Cryptographic Hardware and Embedded Systems*, vol. 2026, no. 2, pp. 736–763, 2026. DOI: [10.46586/tches.v2026.i2.736-763](https://doi.org/10.46586/tches.v2026.i2.736-763).
- Tanvir Hossain**, Ashutosh Ghimire (joint first author), Fathi Amsaad, Tamzidul Hoque, and Ahmed Sherif, “**AI-Assisted Side-Channel Analysis for Golden-Chip Free Detection and Localization of Hardware Trojans**,” *Journal of Hardware and Systems Security*, 2026. DOI: [10.1007/s41635-026-00182-4](https://doi.org/10.1007/s41635-026-00182-4). (Accepted for publication, forthcoming online)
- Sriram Prabhakara Rao, Shree Ram Abayankar Balaji, Farishta Rahman, Tasneem Suha, Tanzim Mahfuz, **Tanvir Hossain**, Mahmudul Hasan, Md Sakib Hasan, Tamzidul Hoque, Prabuddha Chakraborty, and Prakash Ranganathan, “**Secure and Trustworthy Microelectronics: Vulnerabilities, Solutions, and Trends**,” *Journal of Hardware and Systems Security*, vol. 9, pp. 163–189, 2025. DOI: [10.1007/s41635-025-00165-x](https://doi.org/10.1007/s41635-025-00165-x).
- Md Mashfiq Rizvee, Fairuz Shadmani Shishir, **Tanvir Hossain**, Tamzidul Hoque, Domenic Forte, and Sumaiya Shomaji, “**A Persistent Hierarchical Bloom Filter-based Framework for Scalable Authentication and Tracking of ICs**,” *ACM Journal on Emerging Technologies in Computing Systems*, vol. 21, no. 4, pp. 1–22, 2025. DOI: [10.1145/3748650](https://doi.org/10.1145/3748650).
- Tanvir Hossain** and Md Istiaque Rahaman, “**Band gap tuning and variable quantum barrier of armchair graphene nanoribbons by inducing antidote topologies**,” *Journal of Computational Electronics*, vol. 23, no. 1, pp. 32–41, 2024. DOI: [10.1007/s10825-023-02108-7](https://doi.org/10.1007/s10825-023-02108-7).
- Tanvir Hossain**, Md Istiaque Rahaman, and Mahbub Alam, “**Antidote-induced armchair graphene nanoribbon based resonant tunneling diodes**,” *Semiconductor Science and Technology*, vol. 36, no. 7, article 075023, 2021. DOI: [10.1088/1361-6641/ac01fc](https://doi.org/10.1088/1361-6641/ac01fc).

### Conference Proceedings:

- Tanvir Hossain**, S. M. Mojahidul Ahsan, and Tamzidul Hoque, “**SPHINX: A Framework for Security Primitive Hardware Identification and Extraction**,” *GLSVLSI '26: Great Lakes Symposium on VLSI 2026*, Canandaigua, NY, USA, June 2026. DOI: [10.1145/3787109.3815307](https://doi.org/10.1145/3787109.3815307). (Accepted for publication, forthcoming online)

2. Andrea Ramirez-Salgado, Swarup Bhunia, Pavlo Antonenko, Woorin Hwang, Christine Wusylko, Yessy Ambarwati, **Tanvir Hossain**, Tamzidul Hoque, and Rohan Kalavakonda, “**BOARD# 278: NSF IUOE: Empowering Future Engineers. An Inclusive Curriculum for AIoT and Intelligent Embedded Systems,**” *2025 ASEE Annual Conference & Exposition*, 2025. DOI: [10.18260/1-2--55641](https://doi.org/10.18260/1-2--55641).
3. S. M. Mojahidul Ahsan, Muhammad Sakib Shahriar, Mrityika Chowdhury, **Tanvir Hossain**, Md Sakib Hasan, and Tamzidul Hoque, “**Accurate, Yet Scalable: A SPICE-based Design and Optimization Framework for eNVM based Analog In-memory Computing,**” *Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, pp. 1–9, 2024. DOI: [10.1145/3676536.3676827](https://doi.org/10.1145/3676536.3676827).
4. Andrea Ramirez-Salgado, **Tanvir Hossain**, Swarup Bhunia, and Pavlo Antonenko, “**Board 393: Supporting Hardware Engineering Career Choice in First-Year Engineering Students,**” *2024 ASEE Annual Conference & Exposition*, 2024. DOI: [10.18260/1-2--46979](https://doi.org/10.18260/1-2--46979).
5. Ashutosh Ghimire, Fathi Amsaad, **Tanvir Hossain**, Tamzidul Hoque, and Ahmed Sherif, “**FPGA Hardware Trojan Detection: Golden-Free Machine Learning Approach,**” *NAECON 2023 - IEEE National Aerospace and Electronics Conference*, pp. 181–186, 2023. DOI: [10.1109/NAECON58068.2023.10365812](https://doi.org/10.1109/NAECON58068.2023.10365812).
6. Fairuz Shadmani Shishir, Md Mashfiq Rizvee, **Tanvir Hossain**, Tamzidul Hoque, Domenic Forte, and Sumaiya Shomaji, “**A Persistent Hierarchical Bloom Filter-based Framework for Authentication and Tracking of ICs,**” *3rd IEEE International Workshop on Silicon Lifecycle Management*, 2024. DOI: [10.48550/arXiv.2408.16950](https://doi.org/10.48550/arXiv.2408.16950).
7. Andrea Ramirez-Salgado, **Tanvir Hossain**, Tamzidul Hoque, Swarup Bhunia, Mary Jo Koroly, Bradford Davey, and Pavlo Antonenko, “**Board 265: Engaging Students in Exploring Computer Hardware Fundamentals Using FPGA Board Games,**” *2023 ASEE Annual Conference & Exposition*, 2023. DOI: [10.18260/1-2--42715](https://doi.org/10.18260/1-2--42715).
8. **Tanvir Hossain**, S. M. Mojahidul Ahsan, and Tamzidul Hoque, “**Potential and Pitfalls of Multi-valued Logic Circuits for Hardware Security,**” *2023 IEEE 16th Dallas Circuits and Systems Conference (DCAS)*, pp. 1–6, 2023. DOI: [10.1109/DCAS57389.2023.10130261](https://doi.org/10.1109/DCAS57389.2023.10130261).
9. S. M. Mojahidul Ahsan, **Tanvir Hossain**, Md Sakib Hasan, and Tamzidul Hoque, “**Resistive RAM-based PUF: Challenges and Opportunities,**” *2023 IEEE 16th Dallas Circuits and Systems Conference (DCAS)*, pp. 1–6, 2023. DOI: [10.1109/DCAS57389.2023.10130179](https://doi.org/10.1109/DCAS57389.2023.10130179).
10. **Tanvir Hossain**, Md Shyeem Rahman, Md Mushfiqur Rahman, and Ahmed Rabbi Arini Dibbo, “**Performance Analysis of Graphene Nanoribbon Field Effect Transistor (GNRFET) based 6T and 7T SRAMs,**” *2020 23rd International Conference on Computer and Information Technology (ICCIT)*, pp. 1–5, 2020. DOI: [10.1109/ICCIT51783.2020.9392713](https://doi.org/10.1109/ICCIT51783.2020.9392713). (**Best AUST paper award**)

#### Under Review:

1. **Tanvir Hossain**, Abhinav Mahadevan, Jasper Van Woudenberg, Rajesh Velegalati, and Arindam Bhattacharyya, “**Glitch-Lab: A Hardware-in-the-Loop Optimizer for Physical Fault Injection,**” *arXiv preprint*, arXiv:2609.00502 [cs.CR], 2026. DOI: [10.48550/arXiv.2609.00502](https://doi.org/10.48550/arXiv.2609.00502). (Under review)
2. **Tanvir Hossain**, Sudipta Paria, Swarup Bhunia, and Tamzidul Hoque, “**REFORGE: LLM-Assisted RTL Bug Repair with Structured Context and Verification Feedback,**” *Design, Automation and Test in Europe (DATE) Conference 2027*, 2027. (Under review)
3. Ertewaa Alsahayan, Alex Manley, **Tanvir Hossain**, Mohammad Alian, and Tamzidul Hoque, “**Toward Reliable LLM-Assisted Design Space Exploration Under Multi-Performance and Multi-Cost Constraints,**” *Microprocessors and Microsystems: Embedded Hardware Design (MICPRO)*, 2026. (Major revision)
4. Rajesh Velegalati, Jasper Van Woudenberg, **Tanvir Hossain**, Abhinav Mahadevan, and Arindam Bhattacharyya, “**From Structured Search to LLM Operators: Automating Physical Fault-Injection Campaigns,**” *GOMACTech (Government Microcircuit Applications & Critical Technology) Conference*, 2027. (Under review)
5. Ertewaa Saud Alsahayan, **Tanvir Hossain**, and Tamzidul Hoque, “**REMODEL-V: Reliable LLM-Guided Design Space Exploration for Soft-Error-Aware Microarchitectures,**” *GOMACTech (Government Microcircuit Applications & Critical Technology) Conference*, 2027. (Under review)
6. S. M. Mojahidul Ahsan, Mohammad Sakib Shahriar, **Tanvir Hossain**, Md Sakib Hasan, and Tamzidul Hoque, “**SPICECiM-X: An Accurate Circuit-Level Design and Optimization Framework for Analog In-Memory Transformer Inference,**” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2026. (Under review)
7. **Tanvir Hossain**, Mahmudul Hasan, Jackie Ho, Garrett Rose, Tamzidul Hoque, and Md Sakib Hasan, “**PURL: Physically Unclonable and Reconfigurable Logic Locking,**” *Journal of Hardware and Systems Security*, 2026. (Under review)

RESEARCH EXPERIENCE

The University of Kansas

Graduate Research Assistant, Hardware Security (Advisor: Dr. Tamzidul Hoque)

Jan 2022 – Present  
Lawrence, KS

- Conduct research on hardware security spanning side-channel analysis, fault injection, and hardware Trojan detection and mitigation, with results published in IACR TCHES, ACM TODAES, and the Journal of Hardware and Systems Security.
- Develop golden-chip-free and machine-learning-assisted security evaluation methods for integrated circuits, collaborating with researchers across multiple universities.

Keysight Technologies

Research Intern, Machine Learning Security (Applied ML & Fault Injection)

May 2026 – Dec 2026  
Santa Rosa, CA

- Developing model-guided optimization and adaptive-learning workflows for fault-injection parameter tuning, including experiment design, telemetry instrumentation, scalable hyperparameter search, and reliability metrics.
- Delivering a prototype demonstration targeting improved vulnerability-detection coverage over baseline flows.

NSF Research Experience for Undergraduates (REU)

Graduate Research Mentor

Summer 2025

- Mentored four undergraduate researchers in a collaborative NSF REU program between the University of Kansas and Wright State University.

SELECTED RESEARCH PROJECTS

- Trojan-Assisted Meta-Attacks Against Design-for-Security Primitives:** Developed a unified framework to evaluate how hardware Trojans can systematically identify and neutralize Design-for-Security (DfS) primitives in zero-trust microelectronics supply chains. Demonstrated attacks on Ring Oscillator Physically Unclonable Functions (RO-PUFs) and Dynamically Obfuscated Scan Chains (DOSC), enabling authentication bypass, scan-protection removal, and motivating Trojan-aware countermeasures.
- Runtime Protection of Secrets in Untrusted COTS Processors:** Developed HOACS, a software-based framework that uses Residue Number Coding (RNC) to keep sensitive data encoded during execution on commercial off-the-shelf (COTS) processors. Implemented both C-library and LLVM-based transformations to protect cryptographic and arithmetic workloads from hardware-Trojan leakage and side-channel attacks without requiring hardware modification.
- Side-Channel-Driven Security Evaluation of Opaque COTS Hardware:** Proposed a gray-box evaluation framework for assessing unknown COTS devices before deployment when internal design information is unavailable. The approach leverages power and electromagnetic (EM) side-channel measurements to identify anomalous behaviors, guide vulnerability discovery, and support pre-deployment assurance under zero-trust hardware assumptions.
- Gamified Learning Framework for Computer Hardware Fundamentals:** Developed hardware-based games to teach fundamental concepts of computer hardware using Verilog, FPGA, and ESP32 development boards. *Funded by NSF, 2022 to present.*

GRANTS & FUNDING

Contributed as a graduate research assistant to the following federally funded projects.

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|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MARCH 2022 - PRESENT   | <b>Collaborative Research: Gamified Learning of Computing Hardware Fundamentals</b><br>Funding Agency: National Science Foundation, Award Number: 2142696                                                                                                  |
| OCTOBER 2024 - PRESENT | <b>Collaborative Research: Engaging High Schoolers in Integrated Computer Science and Engineering Through Hands-On Experiences with Microelectronics and Artificial Intelligence</b><br>Funding Agency: National Science Foundation, Award Number: 2405375 |
| 2024 - PRESENT         | <b>Collaborative Research: SaTC: CORE: Medium: KIPPER: A Scalable Learning-Guided Hardware IP Protection Platform</b><br>Funding Agency: National Science Foundation, Award Number: 2350364                                                                |
| MARCH 2026 - PRESENT   | <b>Collaborative Research: REU Site: TRUST: Training Research for Undergraduate Students in Secure and Trusted Systems</b><br>Funding Agency: National Science Foundation, Award Number: 2349629                                                           |

## HONORS & AWARDS

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|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAY 2026      | <b>National Science Foundation (NSF) Travel Grant</b><br>IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2026 Symposium)              |
| JANUARY 2024  | <b>First Place (Poster Presentation)</b><br>In the I2S Student Research Symposium                                                                                |
| MAY 2023      | <b>Champion: Microelectronics Security Challenge (IP security)</b><br>IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2023 Symposium) |
| MAY 2023      | <b>National Science Foundation (NSF) Travel Grant</b><br>IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2023 Symposium)              |
| APRIL 2023    | <b>David D. and Mildred H. ROBB Award</b><br>Awarded by the EECS department of the University of Kansas                                                          |
| APRIL 2023    | <b>People's Choice Award (Poster Presentation)</b><br>Among 16 participants in the 2023 Research Showcase by the Graduate Engineering Association of KU          |
| APRIL 2023    | <b>Third Place (Poster Presentation)</b><br>Among 16 participants in the 2023 Research Showcase by the Graduate Engineering Association of KU                    |
| MARCH 2023    | <b>Second Place (Research Showcasing)</b><br>Among 20 participants in the I2S Student Research Symposium                                                         |
| DECEMBER 2022 | <b>Bangladesh Sweden Trust Fund</b> (A travel grant for Graduate study in USA)                                                                                   |
| JUNE 2022     | <b>National Science Foundation (NSF) Travel Grant</b><br>IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2022 Symposium)              |
| MARCH 2021    | <b>Best AUST Paper Award</b><br>The 23rd International Conference on Computer and Information Technology (ICCIT 2020)                                            |
| FALL 2019     | <b>Tuition Waiver Award</b> (Based on the result of the top 8% students)                                                                                         |
| AUGUST 2018   | <b>Runner-up</b><br>Hardware Exhibition, Inter University Software and Hardware Carnival-2018, University of Asia Pacific                                        |
| APRIL 2018    | <b>Second Runner-up</b><br>Project Showcasing of Technival '18, Khulna University of Engineering and Technology (KUET)                                           |
| FEBRUARY 2018 | <b>Second Runner-up</b><br>Hardware of Engenius '18, Organized by AUST EEE Society                                                                               |
| FEBRUARY 2018 | <b>Champion</b><br>Hardware Showcasing of Technomania '18, AUST CSE Society                                                                                      |

## TEACHING & OUTREACH

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### The University of Kansas

Aug 2024 – Dec 2024

*Graduate Teaching Assistant, EECS*

*Lawrence, KS*

- Led lab sessions and graded for undergraduate digital logic and computer architecture, contributing to course materials and exams.
- Guided project-based learning using the MARS (MIPS) and gem5 (x86) simulators.

### The University of Kansas

Fall 2023

*Co-Instructor*

*Lawrence, KS*

- Co-taught “Hands-On Introduction to Computer Hardware using FPGA and Microcontroller,” developing modules with FPGAs, simulations, and the ESP32 Aha development board.

### The University of Kansas

Summer 2023, 2024

*Summer Camp Instructor, EECS*

*Lawrence, KS*

- Designed and ran gamified, hands-on summer camps that teach high-school students computer hardware fundamentals using FPGAs and basic electronics.

## INVITED TALKS & PRESENTATIONS

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|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| JANUARY 2025  | <b>Leakage Unplugged: A Hands-On Tutorial on Power Side-Channel Attacks</b><br>Research Workshop in the I2S Student Research Symposium 2025 |
| FEBRUARY 2024 | <b>Power Side-Channel and Fault Attacks on ASCON and their Countermeasures</b><br>Information Advantage Scholars Visit at I2S, KU           |

## PROFESSIONAL SERVICE

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|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JOURNAL REVIEWER       | SN Computer Science<br>Physica Scripta<br>Journal of Physics: Condensed Matter<br>IEEE Embedded Systems Letters<br>Journal of Hardware and Systems Security                             |
| CONFERENCE REVIEWER    | Technical Program Committee (TPC): IEEE Computer Society Annual Symposium on VLSI (ISVLSI 2026)<br>2026 ASEE Annual Conference & Exposition<br>2024 ASEE Annual Conference & Exposition |
| OTHER RESPONSIBILITIES | Judge: I2S Undergraduate Student Research Showcase                                                                                                                                      |

## TECHNICAL SKILLS

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|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hardware Description Language:</b>   | Verilog, VHDL.                                                                                                                                            |
| <b>FPGA Design Tools:</b>               | Xilinx Vivado, ISE, SDK; Quartus.                                                                                                                         |
| <b>Programming Languages:</b>           | Python, C, MATLAB, Assembly (MIPS).                                                                                                                       |
| <b>Design &amp; Verification Tools:</b> | Cadence Virtuoso, Genus, Encounter; ModelSim; PSpice, HSPICE.                                                                                             |
| <b>Simulation and Analysis Tools:</b>   | gem5, Intel VTune, McPAT.                                                                                                                                 |
| <b>Hardware Platforms:</b>              | <b>FPGA Boards:</b> HaHa Board (Intel MAX10), Altera DE2-115 FPGA, Basys 3 Artix-7, Nexys A7-100T. <b>Embedded Systems:</b> Raspberry Pi, Arduino, STM32. |
| <b>Compilers and Toolchains:</b>        | LLVM.                                                                                                                                                     |
| <b>Side-Channel Analysis:</b>           | Keysight Spyder, Inspector, Pinata, high-precision EM probe; ChipWhisperer Husky; PicoScope; Tektronix DPO70404C Oscilloscope; Analog Discovery 2.        |
| <b>Scripting:</b>                       | Tcl (Tickle), Bash, Git.                                                                                                                                  |

## SIGNIFICANT COURSES

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|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAJOR COURSES IN B.SC.:  | VLSI - I, VLSI - II, Analog Integrated Circuit, Processing and Fabrication Technology, Computer Architecture, Compound Semiconductor and Hetero-Junction Devices.      |
| MAJOR COURSES IN PH.D. : | Intro. Hardware Security, Advance Software Security, Modern Computer Architecture, Machine Learning, Embedded Machine Learning, Advance Data Science, Computer Vision. |